Problem Set-3.
Codes

1] (i) Decimal to BCD.

a) 286 - 0010 1000 0110.
b) 807 - 1000 0000 0111
c) 429.5 - 0100 0010 1001.0101
d) 158.7 - 0001 0101 1000.0111

(ii) Decimal to XS3:

(a) 286 - 0101 1011 1001
(b) 807 - 1011 0011 1010
(c) 429.5 - 0111 0101 1100.1000
(d) 158.7 - 0100 1000 1011.1010

(iii) 2421 Code

(a) 286 - 0010 1110 1100
(b) 807 - 1110 0000 1101
(c) 429.5 - 0100 0010 1111.1011
(d) 158.7 - 0001 1000 1110.1100

6 8 - 9.
2 4 2 1
0000 - 0.
0001 - 1
0010 - 2
⋮

(iv) 5211

(a) 286 - 0011 1110 1010
(b) 807 - 1110 0000 1100
(c) 429.5 - 0111 0011 1111.1000
(d) 158.7 - 0001 1000 1110.1100.

ans

2) a) 846. , b) 679.8, c) 697.5 , d) 860.8.

3] a] 1010 1100 0110. - 793.
10 12 6

(b) 284.6. , c) 697.4, d) 418.5.

4] (i) BCD:

(a)
$$\begin{array}{r} 275 \\ 493 \\ \hline 768 \end{array} \Rightarrow \begin{array}{r} 0010\ 0111\ 0101 \\ 0100\ 1001\ 0011 \\ \hline 0110\ 1\ 0000\ 1000 \\ 0110 \\ \hline 0111\ 0110\ 1000 = 768. \end{array}$$

(bi)
$$\begin{array}{r} 109 \\ 778 \\ \hline 887 \end{array} \Rightarrow \begin{array}{r} 0001\ 0000\ 1001 \\ 0111\ 0111\ 1000 \\ \hline 1000\ 0111\ 1\ 0001 \\ 0110 \\ \hline 1000\ 1000\ 0111 \rightarrow 887. \end{array}$$

(c)
$$\begin{array}{r} 88.7 \\ 265.8 \\ \hline 354.5 \end{array} \Rightarrow \begin{array}{r} 0000\ 1000\ 1000.0111 \\ 0010\ 0110\ 0101.1000 \\ \hline 0010^{1}\ 1110^{1}\ 1101^{1}.1111 \\ 0110\ 0110\ 0110 \\ \hline 0011\ 0101\ 0100.0\ 0101 \\ 3\ 54.5 \end{array}$$

d)

$$\begin{array}{r} 204.6 \\ 185.56 \\ \hline 390.16 \end{array} \Rightarrow \begin{array}{l} 0010\ 0000\ 0100.0110\ 0000 \\ 0001\ 1000\ 0101.0101\ 0110 \\ \hline 0011\ 1000\ 1001.1011\ 0110 \\ \qquad\qquad\qquad\qquad +0110 \\ \hline 0011\ 1000\ 1010.0001\ 0110 \\ \qquad\qquad\qquad +0110 \\ \hline 0011\ 1001\ 0000.0001\ 0110. \end{array}$$

XS3:

(a)

$$\left.\begin{array}{r} 275 \\ 493 \\ \hline 768 \end{array}\right\} \Rightarrow \begin{array}{r} 0101\ \ 1010\ \ 1000 \\ 0111\ \ 1100\ \ 0110 \\ \hline 1101\ (1)\ 0110\ \ 1110 \\ -0011\ \ \ 0011\ \ -0011 \\ \hline 1010\ \ 1001\ \ 1011 \end{array} \rightarrow 768.$$

(b)

$$\left.\begin{array}{r} 109 \\ 778 \\ \hline 887 \end{array}\right\} \Rightarrow \begin{array}{r} 0100\ 0011\ 1100. \\ 1010\ 1010\ 1011 \\ \hline 1110\ 1110\ (1)\ 0111 \\ -0011\ -0011\ +0011 \\ \hline 1011\ 1011\ 1010 = 887. \end{array}$$

c)

$$\begin{array}{r} 204.6 \\ 185.56 \\ \hline 390.16. \end{array} \quad \begin{array}{r} 0101\ 0011\ 0111.1001\ 0011 \\ 0100\ 1011\ 1000.1000\ 1001 \\ \hline 1001\ 1111\ (1)0000\ (1)0001\ 1100 \\ -0011\ -0011\ 0011\ 0011\ -0011 \end{array}$$

Ans. 0110 1100 0011 · 0100 1001

390.16.

5]. BCD:

a) 920
-356
564

⇒

1001 0010 0000
0011 0101 0110

0101 1100 1010
-0110 -0110

0101 0110 100 = 564

(b) 79
-27
52

⇒

0111 1001
0010 0111

0101 0010 = 52.

c) 476.7
-258.9
217.8

0100 0111 0110 . 0111
0010 0101 1000 . 1001

0010 0001 1101 . 1110
-0110 -0110

0010 0001 0111 . 1000 = 217.5

d) 634.6
-328.7
305.9

0110 0011 0100 . 0110
0011 0010 1000 . 0111

0011 0000 1011 . 0111
-0110 -0110

0011 0000 0101 . 1001 -305.9.

(ii) XS3 code:

$$\begin{array}{r} 920 \\ -356 \\ \hline 564 \end{array} \Rightarrow \begin{array}{rrr} 1100 & 0101 & 0011 \\ 0110 & 1000 & 1001 \\ \hline 0101 & 1100 & 1010 \\ +0011 & -0011 & +0011 \\ \hline 1000 & 1001 & 0111 \end{array} = 564.$$

$$\begin{array}{r} 79 \\ -27 \\ \hline 52 \end{array} \Rightarrow \begin{array}{rr} 1010 & 1100 \\ 0101 & 1010 \\ \hline 0101 & 0010 \\ +0011 & +0011 \\ \hline 1000 & 0101 \end{array} = 52.$$

$$\left.\begin{array}{r} 476.7 \\ -258.9 \\ \hline 217.8 \end{array}\right\} \Rightarrow \begin{array}{rrr} 0111 & 1010 & 1001.1010 \\ 0101 & 1000 & 1011.1100 \\ \hline 0010 & 0001 & 1101.1110 \\ +0011 & 0011 & -0011 -0011 \\ \hline 0101 & 0100 & 1010.1011 \end{array} = 217.8.$$

$$\left.\begin{array}{r} 634.6 \\ -328.7 \\ \hline 305.9 \end{array}\right\} \Rightarrow \begin{array}{rrr} 1001 & 0110 & 0111.1001 \\ 0110 & 0101 & 1011.1010 \\ \hline 0011 & 0000 & 1011.1111 \\ +0011 & +0011 & -0011.-0011 \\ \hline 0110 & 0011 & 1000.1100 \end{array} = 305.9.$$

Binary. Gray.

6) a) 8 → 1000 → 1100 (~~gray~~)

b) 23 → 10111 → 11100

c) 96 → 1100000 → 1010000

d) 246 → 11110110 → 10001101

7) Bin → Gray.

a) 1011 → ~~110~~ 1110

b) 101010110101 → 111111101111

c) 110110010 → 101101011

d) 100001 → 110001

8) Gray → Binary.

~~a)~~ ~~(c) 110110010 → 101101011~~

(a) 1111 → 1010.

b) 101110 → 110100

c) 100010110 → 111100100.

d) 11100111 → 10111010.

9) a) 96_{10} (D) → B → G. → 1100000 → 1010000.

b) 45_{16} → 01000101 → 01100111

c) 234_8 → 010011100 → 011010010.

d) 85_{12} → 101_{10} → 1100101 → 1010111

10) 7 bit Hamming. $P_1 P_2 P_4$. → 3bits.

Data: $D_3 D_5 D_6 D_7$ – 4 bits

a) 1100110.

$P_1 P_2 D_3 P_4 D_5 D_6 D_7$.

P_1 check → 1, 3, 5, 7 = 1010 $C_1 = 0$.

P_2 → 2, 3, ~~4~~, 6, 7 → 1010, $C_2 = 0$.

P_4 → 4, 5, 6, 7 → 0110, $C_3 = 0$.

$\boxed{\text{No error.}}$

b) 0011101 → $C_1 = 1, C_2 = 0, C_3 = 1$,

Hence, error in 5th bit.

Corrected code = 0011001.

c) 0111110 → $C_1 = 0, C_2 = 1, C_3 = 1$, 3rd bit is error

0101100

d) 1010111 → $C_1 = 0, C_2 = 1, C_3 = 1$ error in 6th bit.

1010101 → Correct bit.

11) a) 12 bit Hamming code:

$P_1 P_2 D_3 P_4 D_5 D_6 D_7 P_8 D_9 D_{10} D_{11} D_{12}$.

P_1 → 1, 3, 5, 7, 9, 11. = 1

P_2 → ~~2, 3, 5, 6, 8, 9, 11, 12~~. 2, 3, 6, 7, 10, 11 = 1

P_4 : 4 5 6 7 12 = 1

P_8 : ~~8~~ 8, 9, 10, 11, 12. = 0.

Corrected bit: 111101101001.

(b) (i) 110010100110. Remove parity bits for 8 bit word.

01010110.

(ii) 10110101

(iii) 11001101.